

geoai-datacubes: AI-ready multi-mission satellite data cubes for Earth-observation machine learning

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Summary

geoai-datacubes is an open-source Python pipeline that turns raw multi-mission Earth-observation imagery into AI-ready data cubes for machine-learning and deep-learning workflows. From a single configuration block — a region of interest, a time window, a set of bands — it fetches, co-registers, fuses, and tiles imagery from **26 missions** spanning direct sensor observations (Sentinel-2 L1C and L2A (Drusch et al., 2012), Sentinel-1 RTC (Torres et al., 2012), Landsat 8/9 C2 L2 (Roy et al., 2014), NAIP, PlanetScope 4-band and 8-band, MODIS reflectance and land-surface temperature, HLS, ALOS PALSAR, Copernicus DEM at 30 m and 90 m, USGS 3DEP) and derived products (ESA WorldCover (Zanaga et al., 2022), ALOS Forest / Non-Forest, USDA CDL, USGS LCMAP, Impact Observatory annual LULC, JRC Global Surface Water, Hansen Global Forest Change, Chloris biomass). Permission details are in docs/data_layers.md. Four data providers are unified behind one dispatcher — Element 84 Earth Search, Microsoft Planetary Computer, the commercial Planet Orders API, and a direct_http path for non-STAC datasets such as Hansen GFC on Google Cloud Storage. The output is a multi-band GeoTIFF or Zarr cube on a common Universal Transverse Mercator (UTM) grid at a user-specified resolution, ready to be consumed by a PyTorch Dataset that streams training tiles on the fly without ever materialising them to disk. Three self-bootstrapping Colab notebooks ship with the repository as reviewed examples: a *grand tour* of every mission, an *LULC classification* notebook that trains and compares logistic regression, random forest (Breiman, 2001), XGBoost (Chen & Guestrin, 2016) and a lightweight U-Net (Ronneberger et al., 2015) against a conditional NDWI (McFeeters, 1996) spectral-index baseline, and an *integration* notebook that composes geoai-datacubes with the opengeos/geoai modelling library (Wu, 2026) across a multi-AOI Cleveland / Cincinnati / Columbus experiment. A smoke-tests/ folder ships SLURM-or-bash scripts for every mission.

Statement of need

As foundation models and pretrained wrappers for Earth observation become one pip install away, the bottleneck for machine-learning research on satellite imagery is more often the data preparation pipeline than the model itself. A typical workflow stitches together vendor SDKs, manual reprojection between coordinate systems, per-mission cloud masks, ad-hoc nodata handling, band-specific normalisation, and thousands of tile files written to disk. Each step silently degrades the final model: cloud-smeared nodata biases predictions, integer QA bands resampled bilinearly become nonsense, un-normalised DN values collapse CNN training without an error message. geoai-datacubes consolidates these steps into a single configuration-driven pipeline. The library targets graduate students and postdocs entering Earth-observation

43 machine learning, researchers running cross-modal experiments (optical + SAR + DEM +
44 LULC labels), HPC users moving training jobs from laptop to cluster, and instructors teaching
45 applied remote sensing.

46 State of the field

47 geoai-datacubes is complementary to the modelling-breadth geoai package (Wu, 2026),
48 which wraps foundation and task-specific models but handles raster missions one at a time
49 — no multi-mission fusion, per-band normalisation, or spatially-aware splits. Our library fills
50 exactly that data-engineering gap, and a bundled `select_bands` helper plus `BAND_PRESETS`
51 dict lets geoai's loaders consume our cubes without modification; notebook 03 demonstrates
52 the end-to-end hand-off with honest in-distribution $F1 \approx 0.95$ and held-out-city $F1 \approx 0.05$.
53 Beyond geoai: `pystac` (STAC Community, 2024) handles STAC catalogue queries, `rasterio`
54 (Gillies & others, 2023) and `xarray` (Hoyer & Hamman, 2017) handle raster I/O, and `torchgeo`
55 (Stewart et al., 2022) provides PyTorch datasets for pre-existing benchmark cubes and is the
56 lingua franca for the raster-dataset abstraction. None address the per-band-metadata, multi-
57 mission-fusion, spatially-aware-split, or HPC-integration concerns geoai-datacubes exists to
58 solve.

59 Software design

60 Three architectural decisions distinguish geoai-datacubes from generic raster tooling:

- 61 1. **A declarative per-band metadata system (`band_meta`)** pairing every band with a kind
62 (spectral, sar, elevation, temperature, index, categorical, qa) and a normalisation
63 recipe (`linear`, `log_db`, `mean_subtract`, `kelvin_to_celsius_norm`, `divide`, `one_hot`,
64 `passthrough`, ...). Defaults are documented and inspectable; `apply_band_norm` and
65 `get_band_norm` produce ML-ready features by applying those defaults explicitly. A regex
66 fallback infers a sensible recipe for unregistered bands.
- 67 2. **Multi-mission fusion onto a common UTM grid (`fuse_response_tiffs`)** with per-band-
68 correct resampling — bilinear for continuous reflectance and elevation, nearest-neighbour
69 for categorical and QA bands — and provenance preserved in mission-prefixed band
70 descriptions. A fused cube is a single multi-band cloud-optimised GeoTIFF that any
71 `torchgeo`-style PyTorch dataset can consume directly.
- 72 3. **`nan_handling="auto"` with spatially-aware splits.** Per-kind NaN fill (mean for spectral /
73 SAR, biharmonic in-painting for elevation, NN-rounded for categorical, tile-drop on QA
74 NaN) plus four split strategies (`random`, `block`, `stripes`, `regions`) close the train / test
75 leakage hole that random tile splits leave open in spatially-autocorrelated imagery.

76 A `LazyTileDataset` class lets researchers sweep tile size, stride, augmentation, NaN-handling,
77 and split-method choices at training time without ever materialising a tile to disk.

78 Research impact statement

79 The pipeline underpins ongoing research within the authors' group. Hsu and Moortgat use it to
80 fuse Sentinel-2 reflectance with airborne-LIDAR bathymetry and benchmark classical machine
81 learning against deep learning for transferable satellite-derived bathymetry (Hsu & Moortgat,
82 2026). A parallel interpretability line fine-tunes deep networks on Sentinel-2 cubes assembled
83 by the same pipeline and probes how those networks weight individual bands when regressing
84 water depth (Chowdhury, Radhakrishnan, & Subramoni, 2026; Chowdhury, Radhakrishnan,
85 Hsu, et al., 2026). The library is being adopted by additional graduate students in the BuckAI
86 Observatory for water-resource remote sensing and cross-city land-cover-transfer experiments.

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89 of Earth Sciences. The pipeline's foundational Sentinel-1 / Sentinel-2 acquisition and tiling
90 layer was prototyped by Jain and Radhakrishnan in August–October 2025; the multi-mission
91 expansion, ML/DL notebooks, and the present documentation are by Moortgat; review and
92 testing during 2026 by Chowdhury and Hsu. Open data are provided by ESA Copernicus, USGS,
93 and the Copernicus DEM programme; cloud-hosted imagery by Element 84 and Microsoft
94 Planetary Computer; commercial PlanetScope by Planet's Education and Research Program.

95 AI usage disclosure

96 Code development since May 2026 was accelerated by Anthropic's Claude Code (Opus 4.5–4.7
97 and Sonnet 4.5–4.6 models), which assisted with code generation, refactoring, and drafting. All
98 architectural decisions were made by the human authors, who reviewed, edited, and validated
99 every AI-generated artefact before commit.

100 The repository's commit history is unevenly distributed in time, with insertions concentrated
101 in a small number of multi-day windows in December 2025 and June 2026. This reflects
102 a common academic-software pattern: the code was developed for several months on local
103 workstations and HPC accounts before each round of consolidation and public push. The
104 authors are transitioning toward more incremental public development as the project matures
105 and its external user base grows.

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